

CARSON SPENCER MOORE

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EDUCATION

The University of Texas at Austin – Austin, Texas

B.S. in Mechanical Engineering; Minor: Computer Science

December 2027

- GPA: 3.78

EXPERIENCE

SpaceX (Starbase, Texas) - **Booster Integration Engineering Intern**

May 2026 – August 2026

- Responsible Engineer for 13,000+ component assembly sealing Starship Booster's 33 engine feedlines during Cryo Proof
- Re-designed tooling to seal properly with a high-flow Gaseous Oxygen test assembly, a drain manifold for post-test cryo removal, and gas systems ports enabling decay checks and transducer testing with the Checkouts team
- Directed \$800,000 in machined part procurement, running daily vendor calls to ensure timely, print-conforming delivery
- Ran a daily stand-up and authored a daily status email to 80+ stakeholders, driving alignment across organizations
- Ran FEA on max-pressure blow-off load cases to validate structural and sealing performance under Cryo Proof loading
- Oversaw up to 25 technicians during subassembly build and vehicle integration, dispositioning issues on the floor while designing lineside parts storage solutions and physically integrating tooling on-vehicle to resolve fit issues firsthand
- Saved over 1100 production hours per vehicle through design rate improvements, streamlining planning and storage

Longhorn Racing Electric FSAE (Austin, Texas) - **Dynamics Engineer**

August 2023 – Present

- Designed aluminum rear wheel hub assembly for manufacturing and validated the hub with Ansys Finite Element Analysis
- Designed brake rotors and mounts for maximal air clearing and validated rotor structure/temperature tolerance
- Over 1000 hours of CAD experience in SolidWorks and 200 hours of manufacturing experience at UT's machine shop
- Lectured 20 new and returning members, covering tension, compression, and bending fundamentals for structural design
- Simulated dynamic load cases in SKF SimPro Quick to select rear wheel bearings optimized for fatigue life and load capacity
- Utilized SolidWorks Finite Element Analysis on discrete adjustment attachment to validate its structural integrity
- Engineered a carbon fiber push rod and swept control arm assembly to minimize mass, cutting suspension weight by 5 lbs

SpaceX (Starbase, Texas) - **Structures Build Engineering Intern**

January 2025 – August 2025

- Supported over 5000 hours of technician labor, resolving issues on the floor and managing up to 16 technicians at once
- Designed and built a new weld-focused subassembly build cell for a high-pressure propulsion manifold on the Starship vehicle
- Designed fixture tooling to fit up endpoint flanges on tubes using GD&T to hold tolerance across the 9-meter-long weld cell
- Worked with suppliers to coordinate over \$50,000 in purchases to punctually deliver tools lineside on a tight schedule
- Designed a lift tool rated to 1,100 lbs and proof-tested to 2x Working Load Limit for safe handling of propulsion assemblies
- Designed a tool improving work access and integration for aft flap torque frames on the rocket, saving 50 labor hours per vehicle
- Owned 5 retrofitted flight parts through modeling precise as-built CAD assemblies to manufacture off-nominal parts on assembly

SiDi LAB, UT Austin (Austin, Texas) - **Undergraduate Robotics Researcher**

January 2024 – April 2024

- Integrated an infrared motion-capture camera system with a robotic arm and driver software using Python and the NatNet SDK, enabling real-time position tracking for swarm manufacturing research, presented findings in front of UT Research Faculty

Housing Scout (Austin, Texas) - **Real Estate Agent**

April 2024 – April 2026

- Licensed Texas agent specializing in leasing, matching students to suitable housing accommodations near the UT campus

Community Involvements

Neighborhood Longhorns (Austin, Texas) – **Math Tutor**

August 2024 – Present

- Tutored four 6th-grade students weekly for an hour in mathematics supporting a Title I middle school near UT Austin campus
- Contributed to a 23.7% increase in students' STAAR math testing results between the 2025 and 2026 school years

Texas Wranglers (Austin, Texas) – **Alumni Liaison**

September 2023 – Present

- Coordinate member events with alumni and support backend development/outreach for the organization's alumni database
- Appointed by the Vice President of the organization to serve anonymously on the Texas Wranglers Disciplinary Board
- Assisted disabled veterans at an Easterseals gym for 100+ hours by cleaning equipment and helping gymgoers access machines
- 50+ hours of caretaking with On the Moov Student Accessibility Initiative helping care for a disabled UT Austin student

ADDITIONAL

Honors: Glenn and Nancy Gilkey Endowed Scholarship in Mechanical Engineering Recipient, Selected for Tau Beta Pi Engineering Honors Fraternity, Eagle Scout with Bronze Palms

Computer Skills: SOLIDWORKS CAD, SOLIDWORKS CAM, Python, Microsoft Office, Fusion 360, SKF SimPro Quick, MATLAB

Certifications: Machine Shop, 3D Printing, Laser Cutting

Interests: Tennis, endurance running, college football, alto saxophone